

Module 4: The Magnetosphere

Updated 2026-10-05, 20:41:21 UT.

1 Overview

In this module, we discuss the Earth's magnetic field and its extension into near-Earth space, which we call the magnetosphere. The Earth's magnetic field arises due to the rotation of the liquid outer core of the planet. The resulting magnetic field is approximately a dipole, extending into space out to 10-15 Earth radii towards the sun and much farther in the night side. However, at these large distances the magnetosphere is dramatically distorted by the solar wind, so much so that the “magnetotail” extends to hundreds of Earth radii. The magnetosphere contains plasma, so it hosts a number of complex sources and particle populations, including the neutral current sheet, the ring current, and the radiation belts. Although the magnetosphere shields Earth from solar and cosmic radiation, it also controls much of space weather through its complex reconfiguration and perturbations following solar eruptive events.

2 Key Points

- The Earth's Magnetic Field is approximately that of a magnetic dipole
- Beyond about 40,000 km (~ 7 Earth radii), the magnetosphere is heavily distorted by the solar wind
- The magnetosphere traps plasma due to the magnetic bottle effect
- Trapped populations include the plasma sheet, the neutral current sheet, the ring current, and the radiation belts
- Geomagnetic storms are the response of the magnetosphere to solar forcing, either CMEs or CIRs

Reading:

- Required: Pisacane, Chapters 8 & 9, Sections 10.1–10.3
- Optional: Tascione, Chapter 5

3 Theory

3.1 Origin of the Magnetic Field

The magnetic field of Earth originates in the liquid outer core of the Earth (Figure 1). The rotation of the Earth, combined with heat convection in this source, leads to a process known as a dynamo. The very complex [dynamo theory](#) describes the physics of how a rotating, convecting, and electrically conducting fluid creates and maintains a magnetic field over astronomical time scales.

The resulting magnetic field is close to that of a magnetic dipole, which makes it straightforward to describe mathematically. However, to be most accurate, that dipole is tilted with respect to the Earth's axis of rotation, and shifted slightly from the center of the Earth. Once those corrections are made, the dipole model provides an accurate surface magnetic field with less than 10% error, but can be up to 30% in the south Atlantic anomaly (SAA) region. However, at high altitudes in the space environment, the dipole model breaks down because of forcing from the ionosphere and the solar wind, as described later in this module.

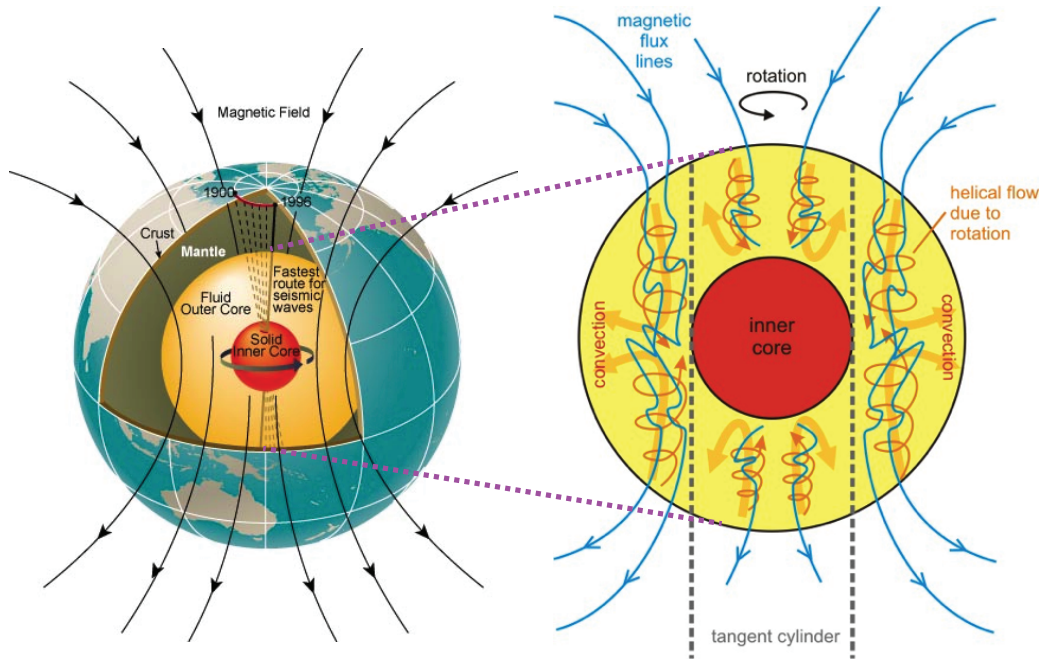


Figure 1: The Earth's core generates a magnetic field through the dynamo action driven by the rotation of the liquid core.

3.2 Dipole Approximation

We wish to describe the Earth's magnetic field with a simple mathematical model that can be used to estimate the expected field at different locations on the surface and in space.

Recall that the **Magnetic Field**, H , has units of Amperes/meter (A/m), while the *Magnetic Flux Density*, B , has units of Tesla (T) or nanoTesla (nT). The two are related through $B = \mu H$, where μ is the **permeability** of the material; in vacuum and in air, the permeability is given by $\mu_0 = 1.257 \times 10^{-6}$ Henrys/meter (H/m). Typically, only magnetic materials have a permeability that deviates from this value.

Many authors define both B and H as the *Magnetic Field*, which can be confusing and annoying. Unfortunately, the standard for the Earth's magnetic field is to use B in nT, but still refer to it as the "magnetic field".

Ampere's Law and the Biot-Savart Law both essentially state that a constant current generates a steady magnetic field. The Biot-Savart Law is most illustrative to derive the dipole magnetic field; it is written as:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_C \frac{I d\mathbf{l} \times \mathbf{r}}{r^3} \quad (1)$$

where I is the current in length element $d\mathbf{l}$, $\mathbf{r}$ is the vector to the observing location, and r is the magnitude of that vector (the distance).

Recall that we model an **electric dipole** as two opposite charges separated by a small distance. Similarly, we model a **magnetic dipole** as a small loop of current with some infinitesimal loop area; this actually gives the same magnetic field as a small bar magnet (Figure 2).

We quantify the magnitude of the source of the magnetic field (the current loop I or the bar magnet) with its **magnetic dipole moment** $\mathbf{m}$, which specifies the strength of the resulting dipole field. The dipole moment $\mathbf{m}$ is a vector quantity that points along the magnet axis, or perpendicular to the current loop. If we align the dipole moment with the vertical axis in spherical coordinates, we can specify the Earth's dipole field as

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \left(\frac{3\mathbf{r}(\mathbf{m} \cdot \mathbf{r})}{r^5} - \frac{\mathbf{m}}{r^3} \right) = \frac{\mu_0}{4\pi} \frac{m}{r^3} (2 \cos \theta \hat{\mathbf{r}} + \sin \theta \hat{\boldsymbol{\theta}}) \quad (2)$$

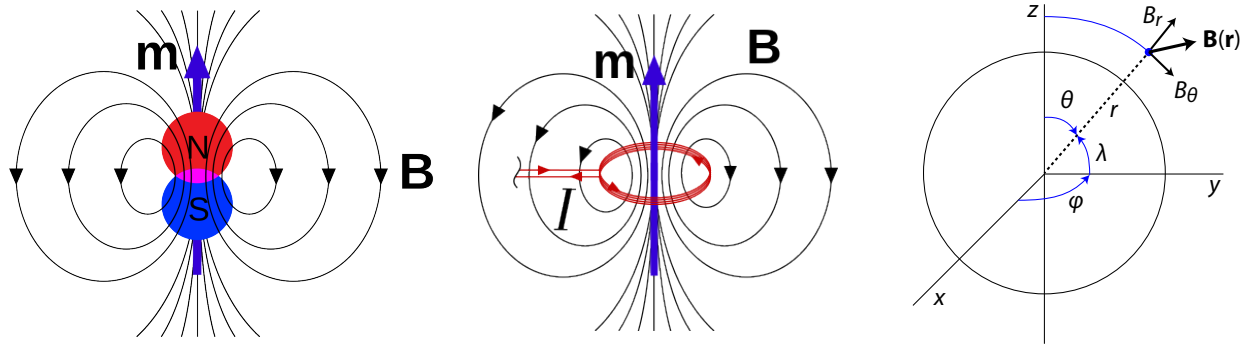


Figure 2: Dipole field generated by a magnet (left) or a current loop (center). m represents the dipole moment. Right: spherical coordinate system for reference.

where m is the magnitude of the magnetic dipole moment. Note that the $\hat{r}$ unit vector points away from the origin, while the $\hat{\theta}$ unit vector points towards away from the vertical axis (Figure 2). The θ direction is often called the *colatitude*; the latitude is given by $\lambda = 90 - \theta$ in degrees. Later in this discussion we will use latitude instead of θ , as a more natural coordinate to describe quantities on the Earth's surface.

Equation (2) shows us that there is a radial component and a horizontal (parallel to Earth's surface) component of the magnetic field:

$$B_r = \frac{\mu_0 m}{4\pi r^3} 2 \cos \theta = -\frac{\mu_0 m}{4\pi r^3} 2 \sin \lambda \quad (3)$$

$$B_\theta = \frac{\mu_0 m}{4\pi r^3} \sin \theta = -\frac{\mu_0 m}{4\pi r^3} \cos \lambda \quad (4)$$

$$\text{or } B_\lambda = -B_\theta = \frac{\mu_0 m}{4\pi r^3} \cos \lambda \quad (5)$$

where we have used the fact that $\cos(90 - \theta) = \sin(\theta)$, and we note that $\hat{\theta}$ and $\hat{\lambda}$ are oppositely directed, hence the minus signs. Note that there is no B_ϕ ; in the dipole approximation, the magnetic field has no azimuthal component (i.e. a component pointing east or west).

If we plug in m for the Earth's magnetic dipole moment we can quantify these equations; however the more common approach is to rewrite the equations as follows:

$$B_r = -2B_0 \left(\frac{R_E}{r} \right)^3 \sin \lambda \quad (6)$$

$$B_\theta = B_0 \left(\frac{R_E}{r} \right)^3 \cos \lambda \quad (7)$$

where R_E is the radius of the Earth. Why write the equations this way? Consider: at the equator, on the surface where $r = R_E$, where $\lambda = 0$, we have $B_r = 0$ and $B_\theta = B_0$; therefore the constant B_0 — which depends only on the constants μ_0 , 4π , and m in the earlier equations, as well as R_E — is precisely the magnetic field strength on the equator. Similarly, at the north pole we have $\lambda = 90^\circ$, so $B_r = -2B_0$ and $B_\theta = 0$; there the magnetic field has magnitude $2B_0$.

The minus sign tells us that the magnetic field at the north pole points *downwards*, toward the center of the Earth; at the south pole it points *upwards*, away from the Earth.

If you want to, you can use these two sets of equations and easily solve for m in terms of B_0 and R_E .

We can combine the first and last of these equations to find the magnitude of the magnetic field anywhere in space:

$$|B| = \sqrt{B_r^2 + B_\theta^2} = B_0 \left(\frac{R_E}{r} \right)^3 \sqrt{1 + 3 \sin^2 \lambda} \quad (8)$$

As before, one can verify from this equation that $|B| = B_0$ at the equator when $r = R_E$ and $|B| = 2B_0$ at the poles.

The basic dipole field, as it extends out into space, is illustrated in Figure 3. The dipole axis is tilted 11.5 degrees with respect to the Earth's spin axis, and that tilt is from the north pole towards 70°W, i.e. over (far) Northern Canada¹. The magnetic axis is also shifted relative the Earth's spin axis, with the shift *away from* the South Atlantic Ocean. For this main reason, the magnetic field above the South Atlantic is somewhat weaker than the rest of the surface field; this is called the **South Atlantic Anomaly** (SAA), and as we will see in the next module, is the cause of higher radiation over this region in low Earth orbit.

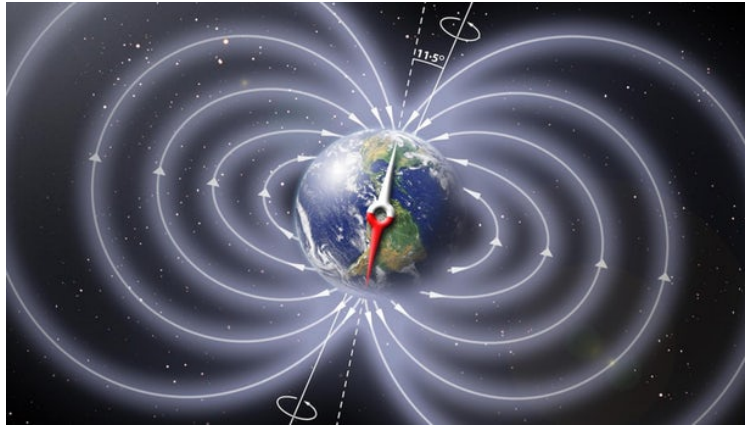


Figure 3: Dipole field of the Earth. The magnetic dipole axis is tilted 11.5 degrees with respect to the Earth's spin axis towards 70°W.

Figure 3 doesn't show the magnitude of the field, but it does show fields lines, which give the direction of the magnetic field. In addition, the **field line density** – that is, the number of field lines in a given area – is directly related to the field strength or magnitude. So, one can consider that the field line density shown in such a plot is, in fact, an illustration of the field strength (though it is not really quantified).

3.2.1 What is a Field Line?

Magnetic field lines are simply a visual tool to show the direction and magnitude of the magnetic field. Since magnetic fields always close on themselves (no magnetic monopoles!), field lines always form closed loops. In the Earth's field, those loops close inside the interior of the Earth (Figure 1).

We can, however, describe a field line mathematically. This is often useful because we want to “trace” field lines in space. A charged particle in the magnetosphere follows magnetic field lines; if my spacecraft measures charged particles at some location in space, we may wish to know where that field line intersects the Earth, so that we can predict where associated events will occur (such as the aurora); or we might want to know where that particle came from, say in the radiation belts.

Mathematically we have

$$\frac{B_\lambda}{B_r} = \frac{rd\lambda}{dr} \quad (9)$$

where dr is the differential element in the r direction, and $rd\lambda$ is the differential element in the λ direction. This equation simply states that the direction of the magnetic field, given by the ratio B_λ/B_r , is equal to the

¹It is a well established fact that Santa Claus has traditionally navigated by compass, since he has been around since long before GPS; and since his home is the north pole, it stands to reason that he actually lives at the magnetic north pole. And since the magnetic north pole is in Canada, this proves that Santa Claus is, in fact, Canadian.

ratio of the incremental distances in each dimension, $r d\lambda$ and dr . We can rearrange and use equations (6) and (7) to find

$$\frac{1}{r} \frac{dr}{d\lambda} = \frac{B_r}{B_\lambda} = 2 \tan \lambda \quad (10)$$

which we then integrate to find:

$$r = r_0 \cos^2 \lambda = LR_E \cos^2 \lambda \quad (11)$$

This is the equation of a field line. Above, r_0 is the distance to the field line at the equator, which is also the largest distance of the field line from the center of the Earth. We have replaced this with $r_0 = LR_E$ to define the distance as a multiple (L) of the Earth's radius; thus we say the field line given by $L = 2$ crosses the equator at $r = 2R_E$.

In space physics nomenclature, values of L actually define not just a field line, but an entire toroidal (donut-shaped) *shell*, or surface, when we wrap the field line around the planet in the azimuthal direction. We therefore refer to these surfaces as ***L*-shells** in the magnetosphere.

3.3 Higher-Order Field Models

The tilted and shifted dipole field model provides an accurate representation of the magnetic field on the Earth's surface with a mean error of less than 10% in the super majority of the Earth's surface, but up to 30% in SAA. However, to close that gap and reduce the error, a more complex model is required.

A *dipole* is a source (electric or magnetic) with two poles. For an electric dipole, it has a positive and negative charge; for a magnetic dipole, think of a bar magnet with a north and south end. To capture higher-order field structure, we need to include higher order “-poles”, such as *quadrupoles* and *hexapoles*².

The easiest way to visualize these quadrupole and higher-order fields is shown in Figure 4. Each of these 3D spherical distributions shows the fields that would result from monopoles (top) down to hexapoles (bottom), noting that there are two angular dimensions on the surface of a sphere. These plots actually show **Spherical Harmonic** basis functions. Spherical harmonics are the 3D, spherical coordinate analogy of the sinusoidal functions used to create a *Fourier series*. In a *spherical harmonic expansion*, we write some function (say, $B(r, \theta, \phi)$) as a linear combination of the spherical harmonics shown in Figure 4.

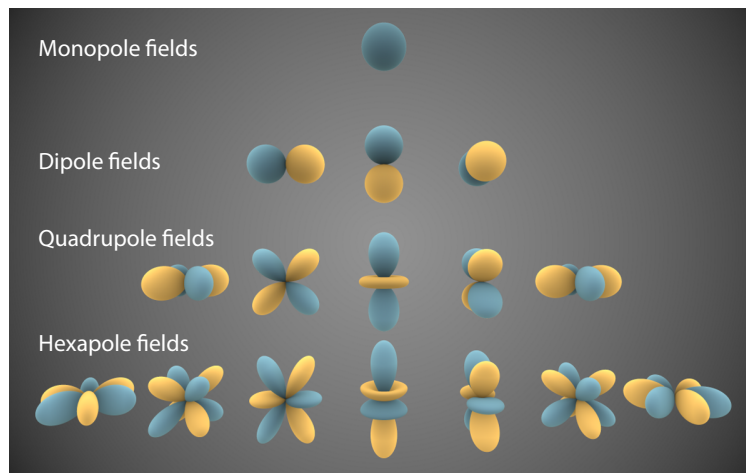


Figure 4: Graphic representation of Spherical Harmonic functions of different orders.

²For electric fields, it is possible to have monopoles and even tripoles: imagine a positive-negative-positive charge arrangement. But for magnetic fields, we can only have even-numbered poles; every north pole has an associated south pole.

Mathematically, the Fourier series expansion of a one-dimensional periodic function $f(x)$ is given by

$$f(x) = a_0 + \sum_{i=1}^N a_i \cos(2\pi n x/T) + b_i \sin(2\pi n x/T) \quad (12)$$

where T is the period of $f(x)$, and we are left to solve for the coefficients a_i and b_i . Analogously, the spherical expansion of a scalar function $U(r, \theta, \phi)$ in 3D spherical coordinates is written as

$$U(r, \theta, \phi) = R_E \sum_{n=1}^N \left[\left(\frac{R_E}{r} \right)^{n+1} \sum_{m=0}^M (g_n^m \cos(m\phi) + h_n^m \sin(m\phi)) SP_n^m(\cos \theta) \right] \quad (13)$$

where R_E is the Earth's radius and $SP_n^m(\cos \theta)$ are specialized polynomial functions known as *Schmidt semi-normalized associated Legendre functions of the first kind*. Don't worry about remembering that! The key to this equation is that just like in the Fourier series, where we need to solve for a_i and b_i , here we need to find the coefficients g_n^m and h_n^m . Note that depending on how many terms we include, we may have many coefficients: if $N = M = 5$, we will have 30 coefficients for each of g and h . An expansion up to $N = M = 14$, typical for the Earth's magnetic field, will have $14 \times 15 = 210$ coefficients each for g and h .

Figure 5 shows the strength of the magnetic field due to each of the spherical harmonic expansion terms. The first term, at $N = 1$ (lower case n on the x-axis of this plot), accounts for the majority ($>90\%$) of the field; it is nearly an order of magnitude stronger than the next term. The first 14 terms describe the B-field produced by the Earth's core (the "core field") very well; higher-order terms start to account for the surface anomalies (crustal fields), with similar magnitudes. It takes hundreds of terms — and thus very high order spherical harmonic functions — to fully account for all of the anomalies.

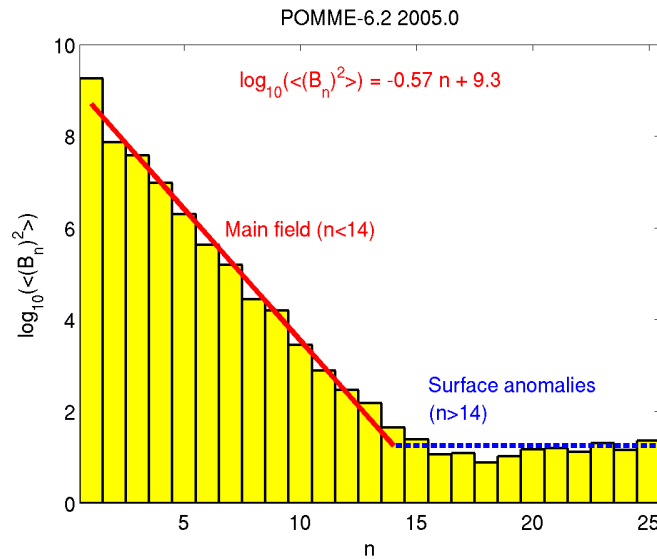


Figure 5: Strength of the B-field produced by each successive harmonic term. The IGRF model goes up to $N = 14$; higher order terms are due to the crustal fields.

The scalar function U above is actually the **magnetic scalar potential**, analogous to the potential V (voltage) for electric fields. It is not a physically-meaningful potential (and we can't measure it), but it is useful for manipulating the math. Once we find the g and h coefficients and write the equation for U , the magnetic field $\mathbf{B}$ is found from

$$\mathbf{B} = -\nabla U \quad (14)$$

Note the similarity between this equation and the relationship between electric field and electric potential, $\mathbf{E} = -\nabla V$.

The dipole field from earlier is a simple solution to the spherical harmonic expansion above if we include only the $n = 1$ and $m = 0$ term. In that case, we have

$$U(r, \theta, \phi) = \frac{R_E^3}{r^2} g_1^0 SP_1^0(\cos \theta) \quad (15)$$

$$= \frac{R_E^3}{r^2} g_1^0 \cos \theta \quad (16)$$

because $SP_1^0(x) = x$ (take my word for it... or look it up on Wikipedia or something). If we now calculate the gradient $\mathbf{B} = -\nabla U$, in spherical coordinates, we will arrive at equations (6) and (7), with $g_1^0 = B_0$; that is, the coefficient of the first term of the spherical harmonic expansion is exactly the dipole field strength at the equator.

Two models are commonly used to fit the Earth's magnetic field to this spherical harmonic expansion: the World Magnetic Model (WMM) and the International Geomagnetic Reference Field (IGRF) model. Both use surface and space-based magnetic field measurements to find the best-fit coefficients g and h that accurately describe the observed field. Once those coefficients are determined, the WMM and IGRF field models are calculated directly using equation (13).

Here's the key takeaway from this discussion about magnetic field modeling. Our descriptions of the magnetic field (i.e., the WMM or IGRF models) are empirically-derived from measurements of the field. However, the resulting field still satisfies the fundamental law of magnetic fields, Gauss's Law, $\nabla \cdot \mathbf{B} = 0$ (Why?).

3.4 The Surface Field

Figure 6 shows representations of the WMM model for 2015. The WMM uses a coordinate system that reports the main field (F), i.e. the magnitude of the field; the inclination (I), not shown in this figure, which is the angle down from vertical; and the declination (D). The WMM is updated every five years, so the most recent update occurred in August 2025.

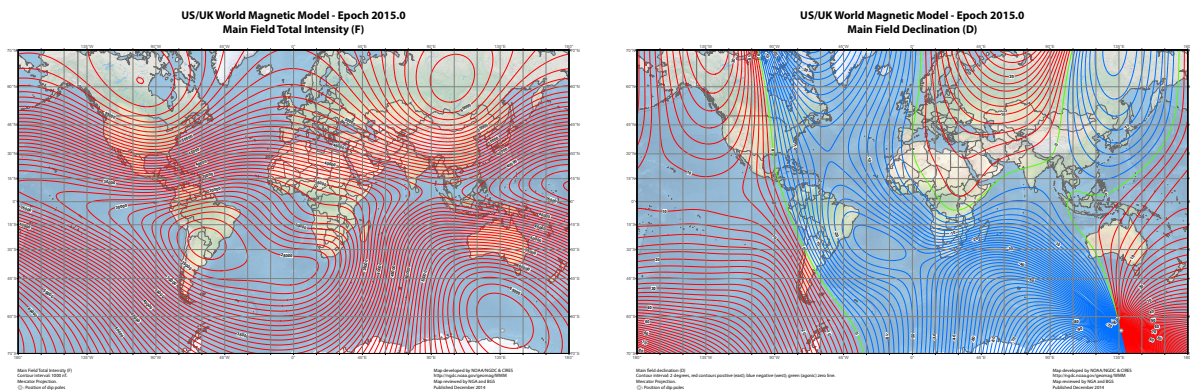


Figure 6: The WMM total field (left) and declination (right).

The left panel of Figure 6 shows the field magnitudes. The magnitude is highest in Northern Canada and in the ocean between Australia and Antarctica; these areas also happen to be where the magnetic poles are located. The field is weakest in South America and over the Atlantic Ocean to the east; this area is the South Atlantic Anomaly mentioned earlier. The weak magnetic field leads to enhanced radiation in this

region, which is a hazard to spacecraft. The contours show the magnitude of the total field. In the SAA, the field reaches a minimum below 25,000 nT; near the poles, it has maxima above 65,000 nT. There is also an anomalously strong field region over Northern Russia and an anomalously weaker field region in the north pacific ocean.

The right panel shows the magnetic declination. The green lines show where the declination is zero; at those locations, a compass needle will point to geographic north. Where the declination is positive (red contours), a compass will point east of north; where it is negative (blue contours), a compass will point west of north. At the poles, the declination is undefined: because the magnetic field is vertical, declination has no meaning. That is why the declination contours all converge at the southern magnetic pole.

Figure 7 shows different components of the International Geomagnetic Reference Field (IGRF). The IGRF model is specified as the Vertical component (Z), the Horizontal component (H), and the Declination (D); from V and H it is easy to calculate both the magnitude (comparable to F) and the inclination I that is also shown, from $\tan I = Z/H$.

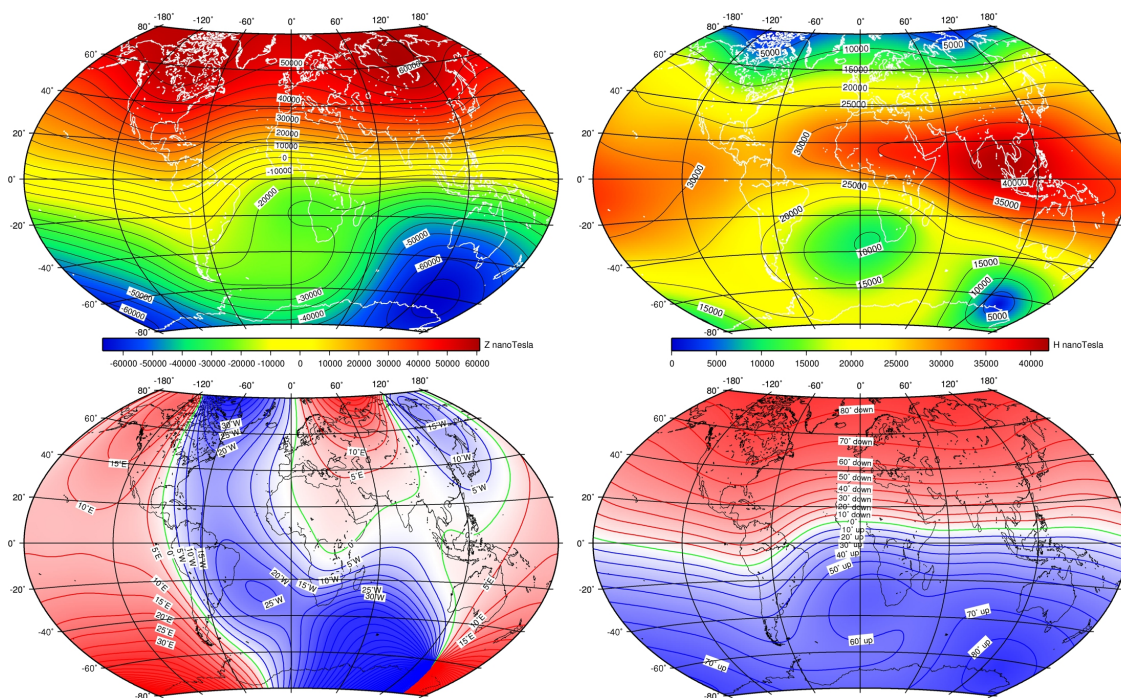


Figure 7: The IGRF model vertical (top left) and horizontal (top right) field components, and the declination (bottom left) and inclination (bottom right).

These figures provide the same information as the WMM model, and a side-by-side comparison shows that they are very similar. The plot of inclination shows one extra feature: the **magnetic equator**, shown as a green line. The magnetic equator is defined as the locations on the map where the inclination angle is zero; that is, the magnetic field is completely horizontal, parallel to the Earth's surface.

The WMM is designed particularly for the surface fields. It is used for navigation by ships and aircraft, and it is used extensively in industry. The WMM is installed on every iPhone and Android phone and automatically corrects the declination of the digital compass reading.

IGRF, on the other hand, provides magnetic fields at altitudes above the Earth's surface with good accuracy. However, it is limited (by design) to the "internal" magnetic field, meaning the field produced by the Earth itself (the core field). It does not include the fields produced by the Earth's crust (crustal anomalies), nor does it account for the "external" field, which is the perturbation to the field produced by

the solar wind and other external forces.

3.4.1 The Changing Magnetic Field

Presumably due to changes in the motion of the inner core, the Earth's magnetic field is changing slowly on timescales of hundreds to thousands of years. In fact, the magnetic field “flips”, with the poles switching places, about every 500,000 years. Evidence for the change in the field can be seen by tracking the location of the magnetic poles. Figure 8 shows the movement of the northern and southern magnetic poles over the last few hundred years. The colored path is the estimated location, based on observations of magnetic materials; the yellow points are actual measurements. The movement of the pole has been acceleration over the past hundred years, for reasons unknown; some speculate that this acceleration is evidence that are approaching the next field “flip”.

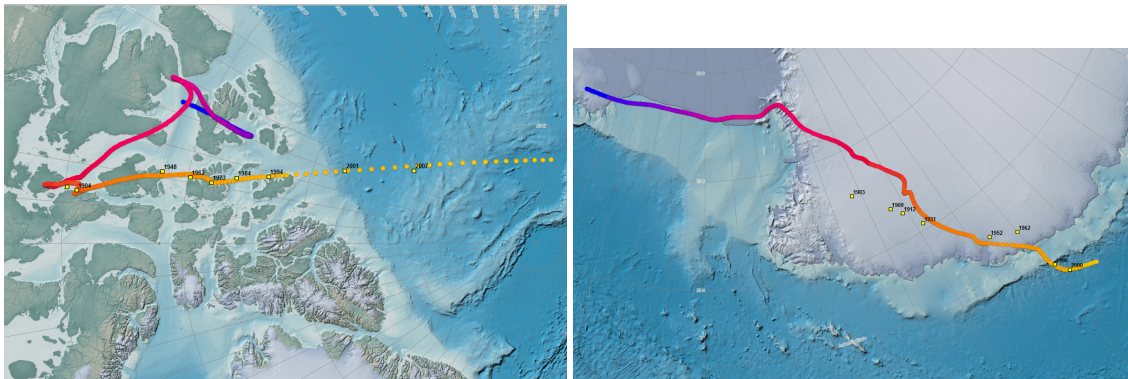


Figure 8: Location of the north (left) and southern (right) magnetic poles, from 1590 (blue) to 2020 (yellow). Measured locations are shown as labeled squares.

3.5 Model of the Magnetosphere Field

The dipole and IGRF models are pretty useful even in near-Earth space, within about six Earth radii; but as we head out further into the magnetosphere, these models start to perform poorly (that is, they don't agree with measurements). Figure 9 compares the dipole field, the IGRF model, and the *Tsyganenko* 2005 (T05) models of the magnetic field. Clearly, the three models agree well inside $L = 6$, with some minor disagreement, mostly on the nightside. At higher L -shells, however, the dipole and IGRF models start to deviate very far from T05.

The *Tsyganenko* models of the magnetosphere have been developed since the 1980s by Nikolai Tsyganenko of the University of St. Petersburg. It has been updated at various times and given labels based on the “released” model: T89, T96, T02, and TS05 (the latter in collaboration with Mikhail Sitnov at APL).

The IGRF and WMM models are based on a couple of assumptions: first, that the field does not change rapidly; and second, that there are no external sources. These assumptions allow us to write equation (13) for the magnetic potential. However, farther out in the magnetosphere, both of the assumptions break down. The magnetic field is full of currents, which we'll discuss later; and it changes rapidly due to forcing by the solar wind.

As such, the only way to model the field is to take lots and lots of measurements and fit them to an empirical model, which now includes time variation due to various solar forcing parameters. The *Tsyganenko* models take measurements of the solar wind density, solar wind speed, and interplanetary magnetic field (IMF), and fit them to a simple function that perturbs the background magnetic field. As more data is collected, the model can be further updated and the coefficients refined.

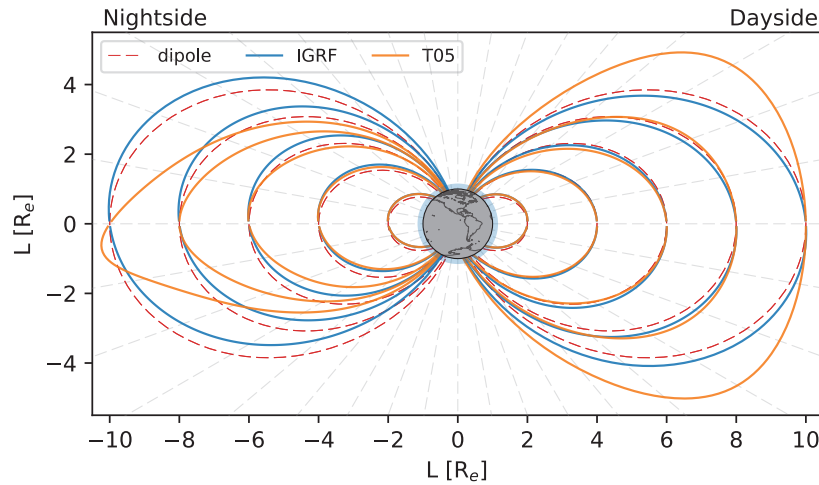


Figure 9: Comparison between the Dipole, IGRF, and Tsyganenko 2005 (T05) field models. Figure courtesy Austin Sousa, CU Boulder.

An example of the T05 field extending out far into the magnetosphere is shown in Figure 10. Notice the wee little Earth at the center; we commonly draw it this way with white and black halves, where the white half is the side facing the sun. Due to the tilt of the Earth's magnetic axis relative the spin axis, the entire magnetosphere is "tilted" near the Earth, but it flattens out in the tail region and points anti-sunward due to the solar wind.

3.6 Magnetospheric Particle Dynamics

The magnetosphere is full of plasma! But before we get further into the magnetosphere and talk about the different plasma populations, we need to learn a bit about how charged particles move in a magnetic field. There are three important particle motions in magnetic fields: i) **gyromotion**, ii) **bounce motion**, and iii) **drift motion**. We'll talk about each of these three in this section, and how they are important in the magnetosphere.

3.6.1 Gyromotion

Recall from Module 1 that a charged particle moving in a magnetic field is subject to the Lorentz force:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}) \quad (17)$$

where $\mathbf{F}$, $\mathbf{v}$ and $\mathbf{B}$ are all vectors. Thanks to the cross product, the force is perpendicular to both the velocity and the magnetic field. Let's assume $\mathbf{B} = B_z \hat{z}$, i.e. the magnetic field is in the z-direction, and the particle is moving in the x-y plane; then we can separate this equation into components (and using $\mathbf{F} = m d\mathbf{v}/dt$, and m is the mass of the charged particle):

$$m \frac{dv_x}{dt} = qv_y B_z \quad (18)$$

$$m \frac{dv_y}{dt} = -qv_x B_z \quad (19)$$

$$m \frac{dv_z}{dt} = 0 \quad (20)$$

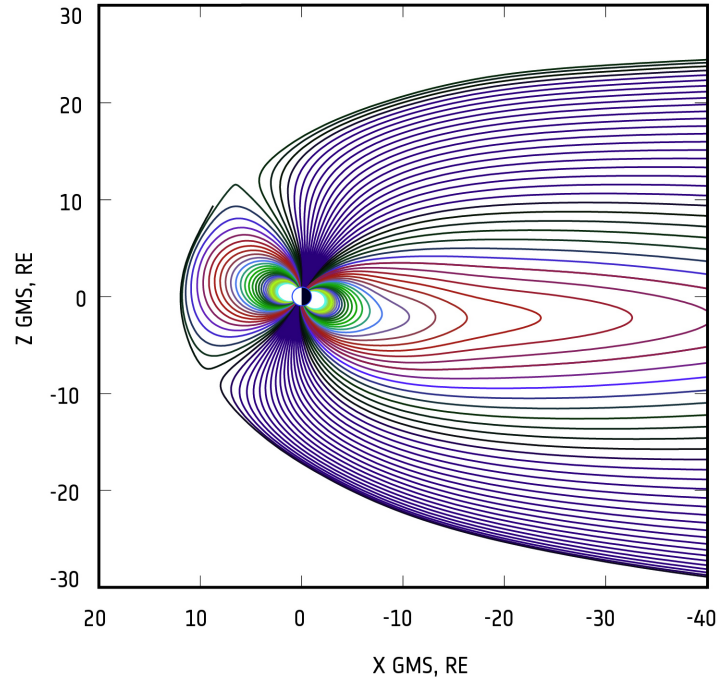


Figure 10: Example of the Tsyganenko 2005 (T05) magnetic field model.

We next take the derivative of the first equation and substitute the second to find:

$$\frac{d^2 v_x}{dt^2} = \frac{q}{m} \frac{dv_y}{dt} B_z = -\frac{q^2 B_z^2}{m^2} v_x \quad (21)$$

This is the equation for a simple harmonic oscillator; the solution is

$$v_x(t) = v_{\perp} \cos(\omega_c t + \phi_0) \quad (22)$$

$$v_y(t) = -v_{\perp} \sin(\omega_c t + \phi_0) \quad (23)$$

where $v_{\perp}$ is the velocity of the particle perpendicular to the magnetic field, and $\omega_c = qB_z/m$ is called the **gyrofrequency** (or cyclotron frequency, hence the subscript c). The two equations above describe a circular motion, perpendicular to the magnetic field, which makes $\omega_c/2\pi$ rotations per second.

How big is the circle drawn out by the particle? The time it takes to make one gyration, the period, is simply $T = 2\pi/\omega_c$. The velocity is $v_{\perp}$ (which obviously depends on the particle energy); then the distance covered, the circumference of the circle, is $C = 2\pi r_g = v_{\perp} T$; therefore the **gyroradius** r_g is given by

$$r_g = \frac{v_{\perp} T}{2\pi} = \frac{v_{\perp}}{\omega_c} = \frac{v_{\perp} m}{q B_z} \quad (24)$$

The gyroradius gets larger for more energetic particles. Notice that the gyrofrequency ω_c is *independent of energy* or velocity of the particle; for a faster particle to make the same number of gyrations per second, it must have a larger gyroradius. On the other hand, a stronger B-field increase the gyrofrequency, and decreases the gyroradius.

Some example values of the gyrofrequency ($f_c = \omega_c/2\pi$) are shown in Table 1 for electrons, protons, and O^+ in different magnetic fields. Thanks to their tiny mass, electrons have very high gyrofrequencies and also very large gyroradii (thanks to fast velocities). Protons have much lower gyrofrequencies, and generally

$ B =$	30,000 nT	3,000 nT	300 nT
f_{ce} (electron)	840 kHz	84 kHz	8.4 kHz
r_g (1 eV electron)	11 cm	1.13 m	11.3 m
r_g (1 keV electron)	3.56 m	35.6 m	355 m
r_g (1 MeV electron)	53.6 m	536 m	5.36 km
f_{cp} (proton)	457 Hz	45.7 Hz	4.57 Hz
r_g (1 eV proton)	2.6 mm	2.6 cm	26 cm m
r_g (1 MeV proton)	2.63 m	26.3 m	263 m
r_g (100 MeV proton)	24.4 m	244 m	2.44 km
f_{cO} (O^+)	28.7 Hz	2.87 Hz	0.287 Hz
r_g (0.1 eV O^+)	0.2 mm	2.1 mm	2.1 cm
r_g (1 eV O^+)	0.7 mm	6.6 mm	6.6 cm

Table 1: Gyrofrequencies and gyroradii for electrons (e) and protons (p) in different magnetic fields

smaller gyroradii, but the gyroradii can become large at very high proton energies. Oxygen ions (O^+) in the ionosphere, at 1 eV and below, are heavy, slow, and gyrate slowly and in tiny circles.

Notice that the gyrofrequency depends of q , which is negative for electrons and positive for protons; that means these particles gyrate in opposite directions. If B points out of the page, *protons gyrate clockwise* and *electrons gyrate counter-clockwise*.

As long as the magnetic field is uniform, a particle will continue to draw the same circle around magnetic field lines indefinitely. However, it is rarely true that the magnetic field is uniform.

3.6.2 Bounce Motion

Above we talked only about the particle motion perpendicular to the magnetic field, $v_\perp$. The particle can also have some component of its velocity parallel to the magnetic field, $v_\parallel = v_z$; however the magnetic field does not affect this velocity, because $v_z \times B_z = 0$. The particle will continue along z unaffected by the magnetic field, while gyrating in the x - y plane; hence it's motion forms a *spiral* around a magnetic field line. The angle of this spiral, α , is given by

$$\tan \alpha = \frac{v_\perp}{v_\parallel} \quad (25)$$

and is referred to as the **pitch angle**. A particle moving in a perfect circle, with no parallel velocity, has a pitch angle $\alpha = 90^\circ$; a particle moving along the field line with $v_\perp = 0$ has $\alpha = 0$. We can now write the perpendicular velocity as $v_\perp = v \sin \alpha$.

A particle moving in a magnetic field has something called a **magnetic moment** μ , which is the ratio of the perpendicular kinetic energy to the strength of the magnetic field:

$$\mu = \frac{mv_\perp^2}{2B} = \text{constant}. \quad (26)$$

One of the key results of plasma physics, known as the *first adiabatic invariant*, says that in a slowly-varying magnetic field, the magnetic moment is constant.

Imagine a particle moving into a stronger magnetic field, as in Figure 11, given as a higher density of magnetic field lines. As the strength of B increases, so must $v_\perp$ in order to keep μ constant above. But the magnetic field cannot provide energy to the particle, or change its speed v , because the Lorentz force is perpendicular to the velocity direction; that means the increase in $v_\perp$ must come at the expense of $v_\parallel$ to maintain a constant speed v .

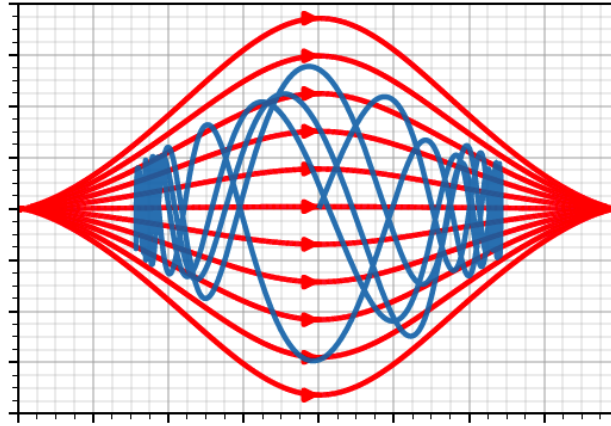


Figure 11: A magnetic bottle. Red lines show the magnetic field; the blue line is the trajectory of a single particle. Notice how it “mirrors” in regions of stronger magnetic field.

Mathematically, we can consider two locations along the particle trajectory with magnetic field strength B_1 and B_2 . The magnetic moment at both locations must be equal:

$$\mu = \frac{mv^2 \sin^2 \alpha_1}{2B_1} = \frac{mv^2 \sin^2 \alpha_2}{2B_2} \rightarrow \frac{\sin^2 \alpha_1}{\sin^2 \alpha_2} = \frac{B_1}{B_2} \quad (27)$$

where α_1 and α_2 are the pitch angles at the two locations. Assume B_2 is stronger than B_1 ; as the particle moves into a stronger magnetic field, its pitch angle *increases*, meaning it's velocity becomes more perpendicular to B .

At some point, B_2 becomes so strong that $\sin \alpha_2$ will reach 90° ; at this point the particle is completely perpendicular to the magnetic field, and it will begin to move the other direction, toward the weaker magnetic field (with pitch angle decreasing). This is known as **magnetic mirroring**. If the magnetic field weakens and then strengthens again at the other end (such as in Figure 11, the particle will continually mirror back and forth between the two points where its pitch angle reaches 90 degrees – it's **mirror points**. Such a magnetic field configuration is referred to as a **magnetic bottle**.

A magnetic bottle is useful for plasma confinement; in the lab, if we create a magnetic field like this, electrons and ions will be confined to this region and won't collide with the chamber walls.

Magnetic field lines in space are magnetic bottles, just curved to stretch from the northern to southern hemisphere of the Earth (Figure 12). Along a single magnetic field line, the field is stronger as we get closer to the Earth, and weakest at the equator; thus a particle will have it's smallest pitch angle at the equator, and will mirror ($\alpha = 90^\circ$) somewhere closer to the Earth. This is how the magnetosphere traps plasma and keeps it confined for long periods of time.

3.6.3 Drift Motion

The Lorentz force equation says that a particle moving with velocity $\mathbf{v}$ in a magnetic field $\mathbf{B}$ will experience a force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$. But it turns out that external forces, F , will influence these particles in interesting ways.

We can see what happens due to an external force F by taking the cross product of both sides of the Lorentz force equation with $\mathbf{B}$:

$$\mathbf{F} \times \mathbf{B} = q(\mathbf{v} \times \mathbf{B}) \times \mathbf{B} \quad (28)$$

But $\mathbf{v} \times \mathbf{B}$ is always perpendicular to $\mathbf{B}$; therefore $(\mathbf{v} \times \mathbf{B}) \times \mathbf{B} = \mathbf{v}B^2$, parallel to $\mathbf{v}$. Substituting this

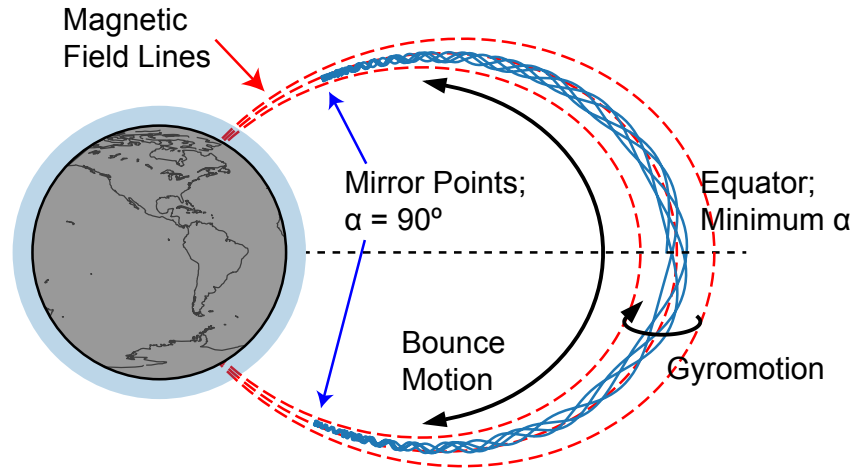


Figure 12: Magnetic Mirroring in the Magnetosphere.

back into the equation above, we have

$$\mathbf{F} \times \mathbf{B} = q\mathbf{v}B^2 \rightarrow \mathbf{v}_d = \frac{\mathbf{F} \times \mathbf{B}}{qB^2} \quad (29)$$

where we have denoted this velocity $\mathbf{v}_d$, the **drift velocity**. This equation says that particles will *drift* at this velocity, in a direction perpendicular to both $\mathbf{F}$ and $\mathbf{B}$.

The easiest way to visualize this is with a proton in an electric field $\mathbf{E}$ that is perpendicular to $\mathbf{B}$ (Figure 13). This is known as $\mathbf{E} \times \mathbf{B}$ (“E cross B”) drift. The force due to the electric field is given by $\mathbf{F} = q\mathbf{E}$; therefore the drift velocity is

$$\mathbf{v}_{dE} = \frac{q\mathbf{E} \times \mathbf{B}}{qB^2} = \frac{\mathbf{E} \times \mathbf{B}}{B^2} \quad (30)$$

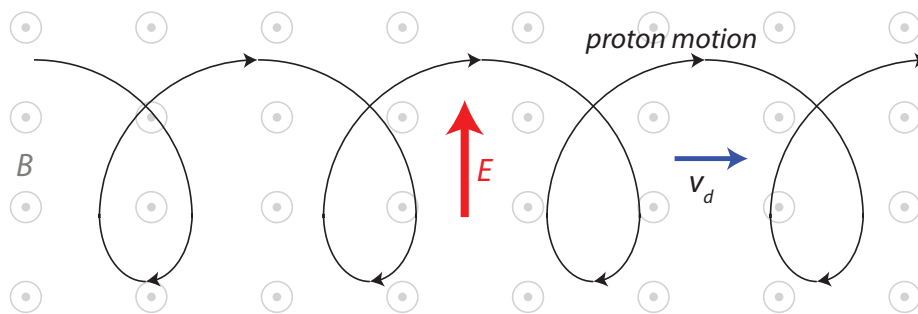


Figure 13: $\mathbf{E} \times \mathbf{B}$ drift.

Remember that the proton gyrates clockwise with the B-field pointed out of the page. As it turns upwards in this picture, it is accelerated by the electric field, increasing its velocity $v_{\perp}$. But increasing the velocity increases the gyroradius (equation 24), so the proton draws a larger circle. As it turns back downwards, it slows down as it opposes the electric field, and its gyroradius is reduced. The result is the motion shown in Figure 13, with a net drift to the right. You can confirm that the right-hand rule gives the drift velocity in the correct direction. Note that the q 's cancelled out above, so the $\mathbf{E} \times \mathbf{B}$ drift does not depend on q ; that means electrons and protons drift in the same direction, and with the same velocity. If there are the same number of protons and electrons, this drift *does not lead to a net current*.

Other forces can cause drifts too; gravity $\mathbf{g}$ causes particles to drift:

$$\mathbf{v}_{dg} = \frac{m\mathbf{g} \times \mathbf{B}}{qB^2} \quad (31)$$

Consider Figure 12; at the equator, $\mathbf{B}$ points upwards and $\mathbf{g}$ points towards the Earth; the drift velocity then points in the East or West direction, around the Earth. This drift depends on mass and charge; electrons and protons will drift in opposite directions due to q , leading to a net current *eastward* (the direction the protons drift). In addition, since protons are ~ 2000 times more massive than electrons, they drift with a much higher velocity, and the electron drift can essentially be ignored as a contribution to the current.

However, the gravity drift is less important than another drift in the same direction, called the **gradient drift**. Recall that the magnetic field varies as $1/r^3$; there is a strong gradient in the magnetic field, which points towards the Earth (towards increasing field strength), the same as $\mathbf{g}$. The force due the magnetic gradient ∇B is

$$\mathbf{F} = -\mu \nabla B = -\frac{1}{2}mv_{\perp}^2 \frac{\nabla B}{B} \quad (32)$$

where μ is the magnetic moment from before. Plugging this into the drift equation, we have

$$\mathbf{v}_{d\nabla} = \frac{mv_{\perp}^2}{2q} \frac{\mathbf{B} \times \nabla B}{B^3} \quad (33)$$

Note that we've eliminated the minus sign by changing the order of the cross product (because $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$). Like the gravity drift, the gradient drift is oppositely directed for protons and electrons (due to the sign of q). It depends on the particle's kinetic energy ($E_{\perp} = \frac{1}{2}mv_{\perp}^2$), so the difference in drift velocity for protons and electrons is not so pronounced (protons do tend to have higher kinetic energy though).

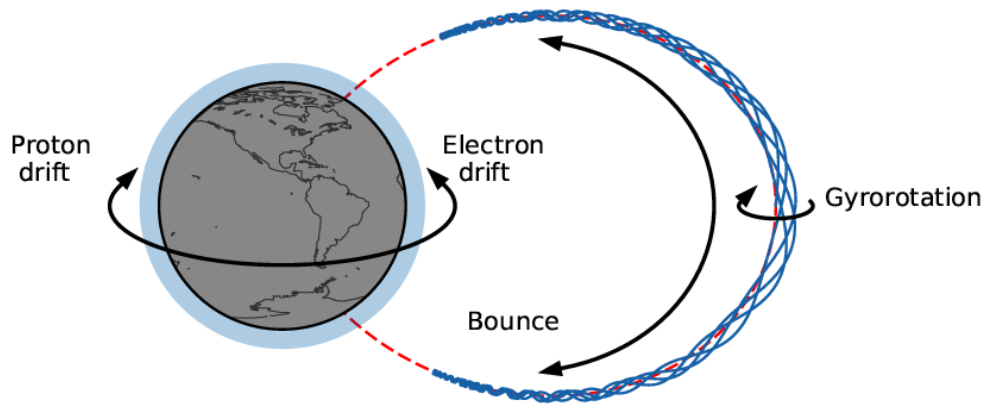


Figure 14: All three particle motions in the magnetosphere: gyromotion, bounce motion, and gradient drift.

Looking at Figure 14 with ∇B pointed toward the Earth and $\mathbf{B}$ pointed up at the equator, the drift velocity for protons will be *westward* (out of the page) and for electrons it will be *eastward*; there is then a net westward current. This is called the **ring current**, and it is extremely important in magnetospheric physics; we'll say more about it in the next section.

The physical intuition of the gradient drift is a bit different than the $\mathbf{E} \times \mathbf{B}$ drift, but it is similarly simple. As a particle makes its circular motion perpendicular to the magnetic field, it enters regions of stronger magnetic field (due to the gradient, which is perpendicular to the field). Since the gyroradius is proportional to $1/B$, the gyroradius gets smaller in this higher B -field. The gyromotion continues back into the region of weaker B , and the gyroradius increases. In this way it creates a similar spirograph motion as shown in Figure 13, where E is replaced by ∇B .

As protons and electrons drift eastward and westward, they will encircle the Earth in some amount of time. This drift motion is essentially how the magnetosphere becomes “filled” in all longitudes.

Just like with the gyromotion, we can estimate the *drift period* of particles, or the time it takes them to circle the Earth. Table 2 gives some examples for protons and electrons at the equator, at different distances from the Earth and different energies, assuming a dipole field model. Note that these times are for protons for different energies; electrons will drift in the same speed if ignoring the relativistic effect. However, if considering the relativistic effect, electrons will drift slower than the protons for the same energy.

$L =$	1.5	3	6
1 keV	488 hours	244 h	122 h
100 keV	4.88 h	2.44 h	1.22 h
1 MeV	29.3 min	14.6 min	7.3 min
100 MeV	17.6 sec	8.8 sec	4.4 sec

Table 2: Drift periods for protons and electrons at the equator in the magnetosphere.

Curvature of the magnetic field (such as along field lines near the equator) leads to another drift called **Curvature drift**. This drift points in the same direction as the gradient drift, thus adding to the ring current. It affects particles as they move along field lines: for a particle to remain on a curved field line, there must be some force allowing its trajectory to curve, and that is the curvature force. We won’t derive the equations here; see Section 6.2 in Knipp, and Table 6-5 and surrounding text in particular.

3.7 Magnetosphere Regions

In this section we come back to the structure of the magnetosphere and qualitatively describe each of its key regions. Figures 15 and 17 provide two equivalent views of the magnetosphere and its regions (there are dozens of similar images online, each of which has its strengths and weaknesses). We’ll start by looking at the first part of the magnetosphere to be hit by the solar wind.

3.7.1 Bow shock, Magnetopause and Magnetosheath

In our discussion of the sun we showed that the solar wind is both supersonic and super-Alfvenic, i.e. the solar wind speed (250–1000 km/s) is much higher than the local sound speed and Alfven speed. When the solar wind, carrying the interplanetary magnetic field (IMF), gets close to the magnetosphere, it begins to slow down; this is because the *magnetic pressure* of the magnetosphere pushes against the *dynamic pressure* of the solar wind. The **Bow Shock** forms where the solar wind speed becomes sub-sonic. This bow shock is about 3–4 R_E in front of the magnetopause, and varies in thickness from 100 km to 2 R_E . Inside the bow shock, the solar wind is subsonic and becomes compressed and heated.

At the magnetopause, by definition, the dynamic pressure of the solar wind is balanced by the magnetic pressure from the magnetosphere. This can be written as:

$$\rho v^2 = \frac{(2B)^2}{2\mu_0} \quad (34)$$

where the extra factor of two arises because the magnetopause current, which cancels the Earth’s magnetic field outside the magnetopause and doubled the magnetic field just inside the magnetopause. This magnetopause current is felt everywhere inside the magnetopause, even on the surface of the Earth. The magnetopause current is formed due to the solar wind plasma gyrating around magnetopause field lines. At the equator, we know that the dipole magnetic field is approximately $B = B_0/L^3$ where L denotes the L-shell, or distance from the center of the Earth in R_E . Using this information, one can calculate the

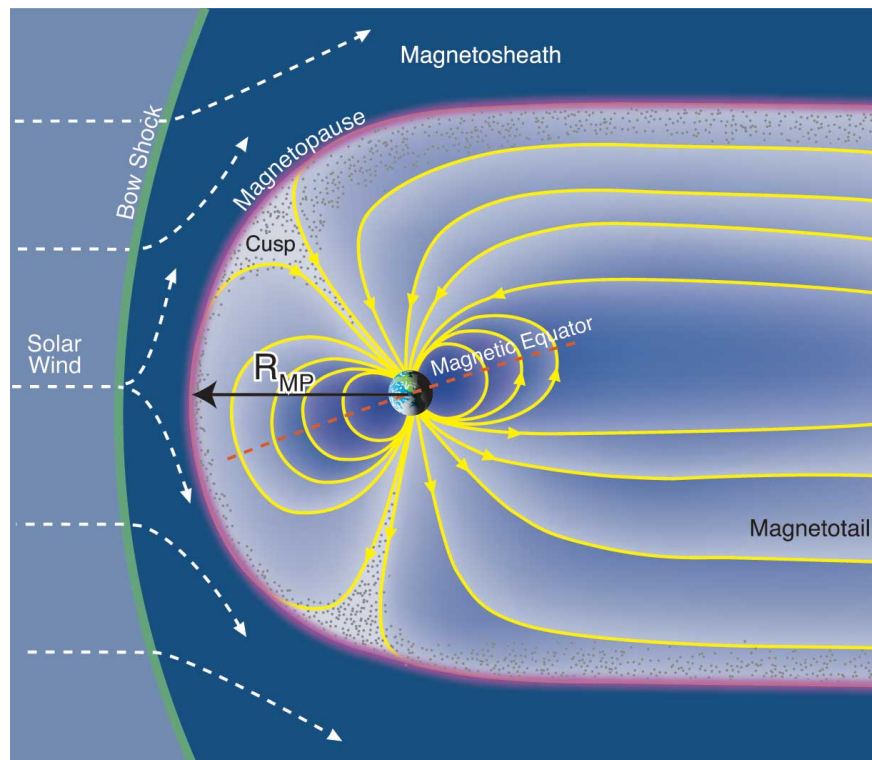


Figure 15: The Magnetosphere.

expected “standoff distance” of the magnetopause, that is, the distance from the center of the Earth to the magnetopause in Earth radii.

Because of the magnetic pressure, very little ($<10\%$) of the solar wind plasma is allowed to enter the magnetosphere. The vast majority is diverted and remains in the **magnetosheath**, the region between the bow shock and the magnetopause. In the magnetosheath, particles are deflected around the magnetosphere, as shown in Figure 15. The density of plasma in the magnetosheath is high compared to the solar wind and just inside the magnetosphere, except for the situation when the plume of plasmasphere extends to the magnetopause.

Because plasma is flowing tangentially just outside the magnetopause in the magnetosheath, but the plasma inside the magnetopause is relatively stationary, there is a shear velocity at this boundary. This shear velocity can lead to *Kelvin-Helmholtz instability* along the magnetopause, with its characteristic “wave” structures (Figure 16). These instabilities create a path for plasma to enter the magnetosphere on the dawn and dusk flanks of the magnetopause.

3.7.2 Polar Cusp

The magnetosheath can be thought of as a region of plasma that is diverted around the magnetosphere. However, there is a direct connection between the magnetosheath (and solar wind) and the Earth at the regions known as the *polar cusps* (Figures 15 and 17). The magnetic field lines in this region of the magnetosphere are nearly vertical and connect to the magnetopause at near normal angles. The cusp regions separate the dayside magnetosphere from the nightside magnetotail region.

Where the cusp field lines meet the magnetopause, the field can be weak and open (‘open’ means the field lines have one end rooted to Earth’s high latitudes and the other end extending out into the solar wind, not returning back to Earth), and so it is relatively easy for plasma in the magnetosheath to funnel into the

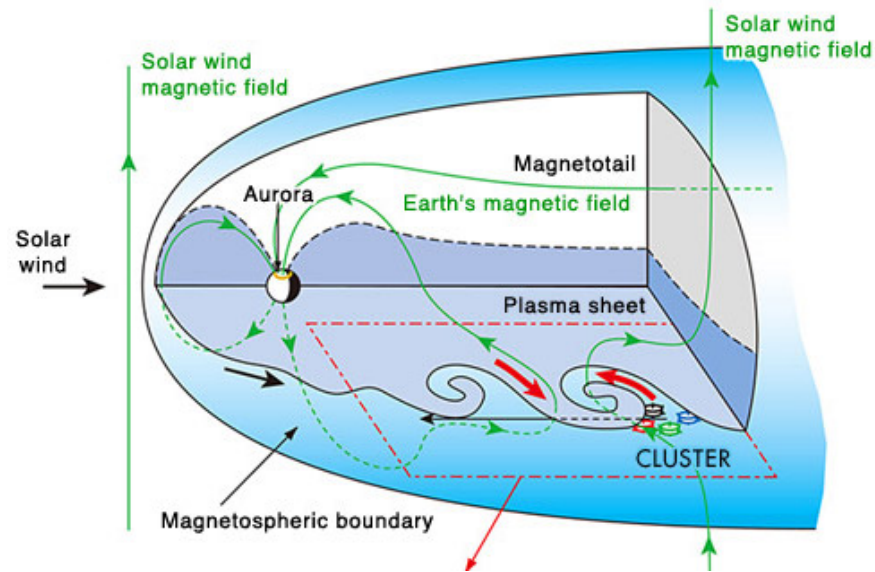


Figure 16: Kelvin-Helmholtz instability along the magnetopause, created by the shear velocity of the plasma in the magnetosheath.

cusps. These particles can then follow the field lines all the way down to the Earth's atmosphere. Fluxes of high-speed particles bombard the upper atmosphere in the cusp regions which occur near 75° geomagnetic latitude.

3.7.3 Plasmasphere and Plasma sheet

Recall from our discussion of the ionosphere that the **plasmasphere** is basically the extension of the ionosphere out into the magnetosphere. The boundary between the topside ionosphere and plasmasphere, which occurs at about 800-1000 km altitude, is the location where thermal pressure and magnetic pressure are equal. In the plasmasphere, the plasma is dominated by magnetic pressure, and therefore becomes confined by the magnetic field. On the other hand, the ionosphere is dominated by thermal pressure, and so is able to cross magnetic field lines to cover the entire planet.

The plasmasphere plasma is *cold*, with characteristic energies of 0.1–1 eV (temperature $\sim 10^4$ K). The density in the plasmasphere falls off slowly within a few Earth radii, and then drops precipitously at the **plasmopause**. The plasmopause location varies with geomagnetic activity, extending to the dayside magnetopause during extended quiet times and being pushed in closer to the Earth during geomagnetic storms.

Beyond the plasmopause lies the **plasma sheet**, a large region of space that extends far into the magnetotail. The plasma in this region is relatively hot, with electron temperatures of 2–20 million Kelvins (5–7 times higher for protons). Major differences between the plasmasphere and plasmasheet are:

- The plasmasphere is cold; the plasmasheet is hot
- The plasmasphere is relatively dense ($10\text{--}1000$ ions/cm³); the plasmasheet is much more tenuous ($0.1\text{--}1$ ions/cm³)
- The plasmasphere is confined to the Earth's magnetic field, and partially corotates with the Earth (there is some lag). The plasmasheet does not corotate with the Earth.
- The plasmasphere is sourced by ions and electrons from the ionosphere; the plasmasheet is a combination of ionosphere and solar wind plasma.

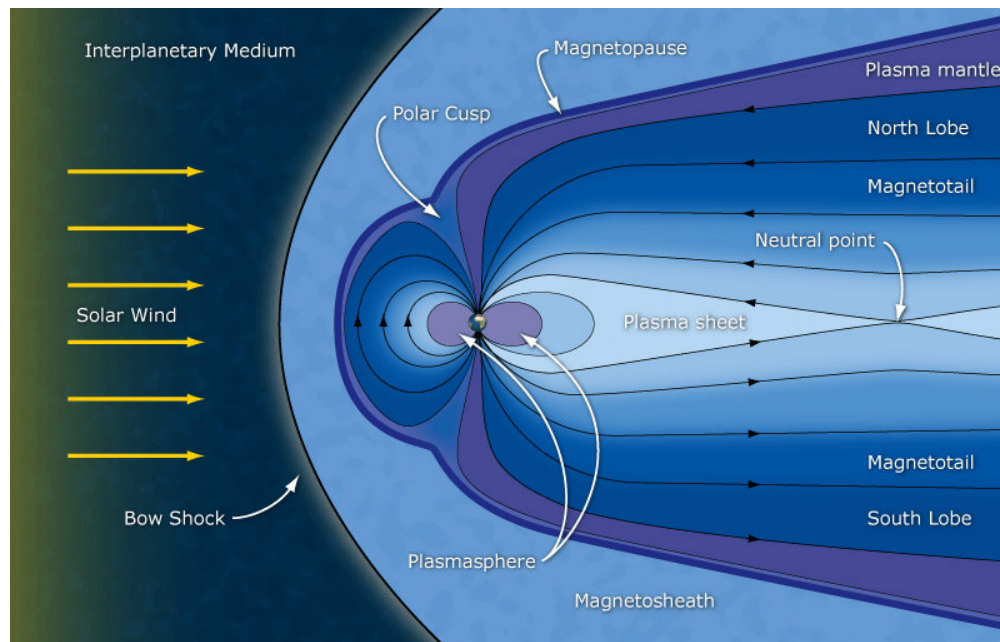


Figure 17: The Magnetosphere, version 2.

The plasmasheet exists in a region of space that magnetically “maps” to the auroral region of the Earth’s surface: the **auroral oval**. As such, processes occurring in the plasmasheet during geomagnetic storms and substorms lead to auroral activity in the upper atmosphere.

3.7.4 The Ring Current

Earlier we talked about the *gradient drift* (and alluded to the curvature drift) which leads to a current traveling westward around the planet (protons drift westward, electrons drift eastward). The gradient drift is proportional to particle energy ($\frac{1}{2}mv_{\perp}^2$) and proportional to $1/r^4$ (plug in for B and ∇B to see this); as such, there is a strong current in the plasmasheet, where particle energies are higher and r is larger, compared to the plasmasphere.

This **Ring Current** is one of the most important features of the magnetosphere. It is composed of high-energy electrons and ions with energies between 10–300 keV, with an average energy of 85 keV (the average can help you find the average current density of about 10^{-8} A/m²). It is confined mostly to 3–6 R_E , at the boundary of the plasmasphere and plasmasheet, and is about 4–5 R_E thick (in the north-south direction).

The Ring Current is even hotter, but less dense, than the plasmasheet where it overlaps. Figure 18 compares the temperatures and densities of different regions of magnetospheric plasma.

3.7.5 Other Magnetospheric Regions

A few other regions are evident in Figure 17 that we haven’t mentioned yet. We won’t do a comprehensive study of these regions, but here we’ll provide a basic, qualitative description.

The Plasma Mantle is the region of the magnetosphere tailward of the cusps, nearest the magnetopause. This region acts as a barrier to the solar wind, although some solar wind particles become trapped in a layer of the mantle just a few kilometers thick. The mantle is a region of enhanced magnetic field due to the interaction with the solar wind magnetic field. The region is realistically much thinner than shown in Figure 17.

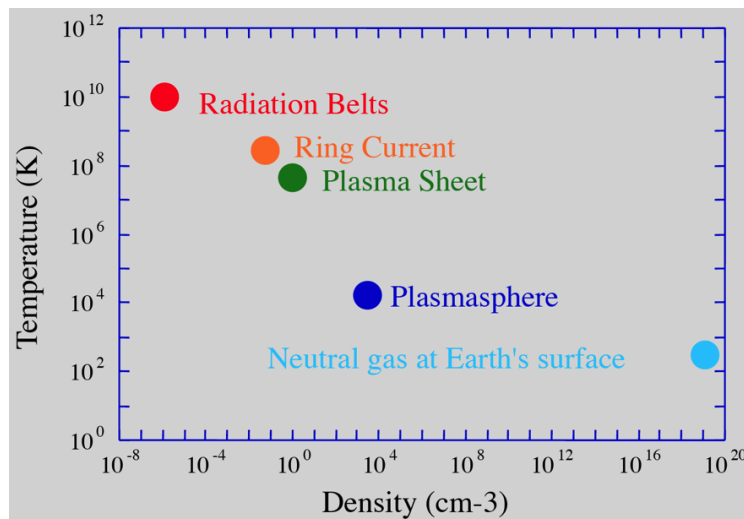


Figure 18: Temperatures and densities of different regions of magnetospheric plasma.

The Lobes are the regions just inside the mantle. These regions contain only a small amount of plasma originating from the ionosphere. The magnetic field lines in the lobes map directly to the polar regions of the Earth. In the lobes, the magnetic field is strong enough to create enough magnetic pressure to sustain the elongated magnetosphere, preventing it from collapsing on itself back into a more dipole-like configuration.

The Magnetotail resides between the lobes and the plasmasheet. It stretches to hundreds of Earth radii; very few spacecraft have been launched into orbits to directly investigate this region. Because the field itself becomes very weak with distance, and the density is also very low, the magnetotail becomes wispy.

Deep in the magnetotail is the second location where solar wind field lines and magnetosphere field lines “reconnect” when the solar wind magnetic field points southward. This occurs somewhere between 25–150 R_E . When reconnection occurs in the distant magnetotail, compressed regions of plasma called *plasmoids* can be accelerated towards the Earth and away from the Earth on either side of the reconnection point. We’ll say more about this in the section on geomagnetic storms.

3.7.6 Other Magnetospheric Currents

In addition to the ring current, there are a number of other current systems that exist in the magnetosphere which control the dynamics of the space environment. Here we provide just a brief definition of each of these:

- **Magnetopause Currents:** Currents flow along the magnetopause boundary, generally perpendicular to the magnetic field direction, thanks to the Lorentz force. On the day side these currents generally flow west-to-east near the equator, as shown in Figure 19, while the tail magnetopause currents flow along the top and bottom boundaries of the magnetosphere.
- **Neutral Current Sheet:** In the flattened region of the plasmasheet far in the magnetotail (Figure 17), the magnetic fields nearly cancel, making the region magnetically “neutral”. A thin current sheet flows east-to-west in this region.
- **Region 1 and Region 2 Field Aligned Currents:** Currents carried mostly by electrons along magnetic field lines from the magnetopause and plasmasheet to the ionosphere. These currents are driven

by electric fields and are mostly carried by electrons. The two currents (Region 1 and Region 2) flow in opposite directions, and close in the ionosphere in the *cross-polar-cap current* or *Pederson current* (Figure 19).

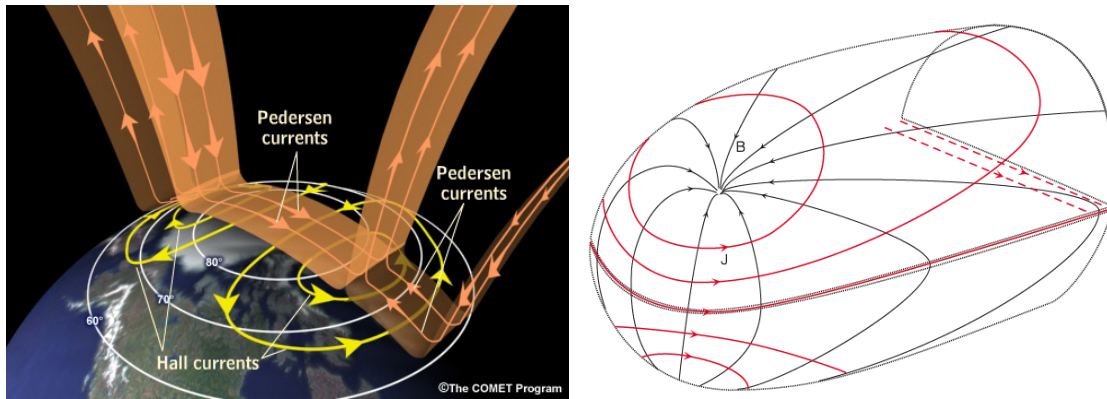


Figure 19: Left: Field-aligned currents and their closure in the ionosphere. Region 1 currents are closer to the poles; Region 2 currents are closer to the equator. Right: magnetopause currents shown in red flowing along the magnetopause. The Neutral Current Sheet is also shown as a dashed red line.

4 Effects

4.1 Geomagnetic Storms

Geomagnetic storms are defined as multi-day disturbances in the geomagnetic field. However, they are commonly observed as a disturbance of the magnetic field, measured by ground stations near the equator on the Earth's surface, but the effects of geomagnetic storms are much more broad. Geomagnetic storms are driven by either coronal mass ejections (CMEs) or corotating interaction regions (CIRs), both of which are enhanced plasma structures in the solar wind, which bring with them the perturbed solar wind magnetic field.

Geomagnetic storms are typically measured using a handful of magnetometer stations near the equator (Figure 20), but purposely not at the equator to reduce the effects of equatorial electron jets, which were discussed in the previous Module (Ionosphere). These stations measure the change in the horizontal magnetic field, in units of nanoTesla (nT). This perturbation is known as the *disturbance storm-time* or **Dst index**. Figure 20 shows an example for the 2024 Mother's Day³ storm, which observed a Dst of -400 nT. Dst index from -50 to -100 nT is considered a "moderate storm"; from -100 to -250 nT is an "intense storm"; and below -250 nT is a "super storm" or "great storm".

The Dst is negative during a geomagnetic storm thanks to the ring current. As described above, the ring current flows westward around the Earth. During a storm, the ring current is enhanced; this current then leads to a downward magnetic field at the Earth's surface, which is added to the quiet, background magnetic field. Thus, the Dst magnetic field perturbation is actually a remote measurement of the enhancement of the ring current, which is the dominant current for the Dst index. Other currents, such as the magnetopause current and magneto-tail current also contribute to the Dst.

Geomagnetic storms are measured in other, similar indices as well. The *Kp index* is the planetary average of the K index, which is itself a semi-logarithmic measure of the disturbance of the horizontal magnetic field at the Earth's surface. The biggest difference is that the K index stations are at subauroral latitudes (50 – 60°

³We have a tradition of naming geomagnetic storms after the nearby holidays, for reasons that I do not know.

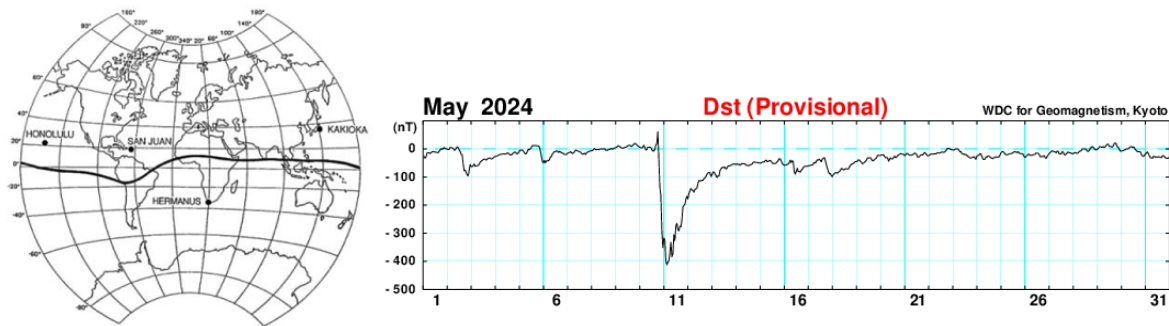


Figure 20: Left: Map of Dst monitoring stations. Right: Example Dst trace for May 2024; the big drop late on May 10, 2024 is the now-famous “Mother’s Day Storm”.

North), while Dst is measured near the equator. The K index and Kp index ranges from 0 to 9, and also includes intermediate values (0+, 1-, etc.). Each unit step (i.e. from 1 to 2) is approximately a factor of 2 increase in magnetic activity.

The linear equivalent of the Kp index is the **ap index** (sometimes written with a capital A, sometimes not).

4.1.1 Geomagnetic Storm Development

A geomagnetic storm begins with the *Initial Phase*, when the CME or CIR plasma structure arrives at the bow shock. The magnetosphere becomes compressed due to the enhanced solar wind dynamic pressure; there is then a small positive change in the Dst. Now, if the interplanetary magnetic field (IMF) points northward, there is no significant energy from solar wind to be parted into the magnetosphere; but if the IMF points **southward**, magnetic reconnection allows solar wind plasma to enter the magnetosphere and much more of the solar wind energy will be parted in to the magnetosphere, and this begins a *sudden storm commencement (SSC)*.

The *Main Phase* of a geomagnetic storm lasts for several hours to one day. The reconfiguration of the magnetosphere and increased magnetosphere plasma leads to an enhancement of the ring current, which also moves closer to the Earth. The enhanced ring current drives a big negative change in Dst to its minimum value. During the storm main phase, the magnetosphere is rapidly reconfiguring, and the magnetic and electric fields and the plasma are attempting to find a new equilibrium.

In the *Recovery phase* of a storm, all of the new energy stored in the magnetosphere is slowly dissipated. Electrons and ions are lost to the atmosphere (through precipitation) and the magnetopause; wave-particle interactions contribute in moving plasma. The recovery phase can be punctuated by impulsive substorms, driven by smaller changes in the IMF direction. The recovery can last for days to several days. It is also common for storms to come in pairs, separated by a day or two; these are observed as a double-dip in the Dst index. For example, the famous “Halloween storm” in 2003 included two storms, almost exactly 24 hours apart, both of which reached $Dst < -350$ nT.

4.1.2 Geomagnetic Storm Effects

Geomagnetic storms have quite a number of effects that are observed on the ground (Dst index) and by spacecraft:

- The magnetosphere gets compressed and the magnetopause moves inward

- If the IMF points southward, reconnection occurs on the dayside and in the tail. Field lines convect rapidly around the planet.
- Solar wind plasma gets into the magnetosphere
- The ring current can be enhanced by a factor of 10 or more. This enhancement leads to a reduction of the magnetic field at the Earth's surface, measured by the Dst index
- The plasmasphere *shrinks*, a better description is that plasmasphere becomes eroded due to enhanced convection that transports plasma from the outer plasmasphere toward the dayside magnetopause, and the plasmapause moves inward.
- The radiation belts are normally enhanced, by a factor of 10 or more
- Auroras occur at high latitudes; for larger storms, auroras can be driven to lower latitudes due to the reshaping of the magnetotail into a more dipole-like structure

4.1.3 Geomagnetic Substorms

Contrary to what the name suggests, *substorms* are not necessarily associated with storms. While substorms can occur during the recovery phase of geomagnetic storms, they can also occur in isolation.

Substorms can occur during the quiescent solar wind – they are not driven by CMEs or CIRs. There can be unsteady solar wind with fluctuations in the direction of the IMF, which drive similar (but smaller) geomagnetic effects. Substorms occur 3–4 times per day. They drive magnetic field convection around the planet, which in turn leads to the release of stored energy in the magnetotail, leading to auroral precipitation. These substorms are measured by their auroral observations in the absence of large Dst perturbations, but are associated with magnetic perturbations at high latitude (i.e., the AE (Auroral Electrojet) index).

Both Dst and AE (AL and AU) indices are forecast in real-time based on empirical models developed at Prof. Li's research group, using the real-time solar wind measurements at L1, check out the following website: https://lasp.colorado.edu/space_weather/dsttemerin/dsttemerin.html

4.2 The Aurora

The Aurora Borealis (Northern Lights) and Aurora Australis (Southern Lights), or just the aurora, are produced when energetic electrons (1–50 keV and sometimes higher) are accelerated down magnetic field lines at high latitudes and strike the atmosphere. The aurora are the first direct evidence of space weather, and obviously have been observed for thousands of years: really for as long as humans have lived at high latitudes.

As mentioned earlier, the aurora occurs at latitudes that connect to the plasmasheet. When a geomagnetic storm arrives, the magnetosphere is perturbed and field lines convect around the Earth to the magnetotail. In the tail, magnetic reconnection occurs, and the earth side of the field lines convect towards the Earth and carry plasma with them. The magnetic field configuration becomes more dipole-like and plasma can be accelerated along field lines where it collides with the upper atmosphere.

When these energetic electrons collide with atmospheric molecules, they deposit energy in three ways: i) excitation, ii) dissociation, and iii) ionization. The excitation of atmospheric O and N₂ leads to the optical emissions that we think of as the aurora; Atomic oxygen (O) emissions are green at 557.7 nm wavelength (green) at 100 - 150 km altitude and a wavelength of 630 nm (red) at 200-300 km. These are the main contribution to the auroras, while N₂ emissions occur over a wide range of wavelengths, from ultraviolet to infrared (the red and purple aurora are the visible N₂ aurora) as minor contribution.

The aurora comes in many forms and structures. There are “diffuse” aurora which are relatively boring to view, all the way through “breakup” events, which are dynamic on second time scales and absolutely stunning to watch. There are “flickering” and “pulsating” aurora, and there are “black” aurora (bands absent of aurora within diffuse or pulsating aurora). All of these different forms are driven by different processes

in the magnetosphere, involving the magnetic fields, particle populations, and electromagnetic and plasma waves.

This should answer the next question: what are they good for? Apart from a tourist attraction, the aurora are an invaluable tool to study space physics. Researchers around the globe use auroral observations – often in conjunction with spacecraft measurements – to understand the magnetic fields, currents, and waves that are driving the dynamic phenomenon observed in the upper atmosphere. With conjunction measurements on satellites, the discrete auroral features also provide a method of magnetic field mapping, showing where field lines in the magnetosphere connect to the atmosphere. But that relies on our ability to directly connect the phenomena we observe in the magnetosphere (plasma waves, particle populations, etc) with the aurora they produce!

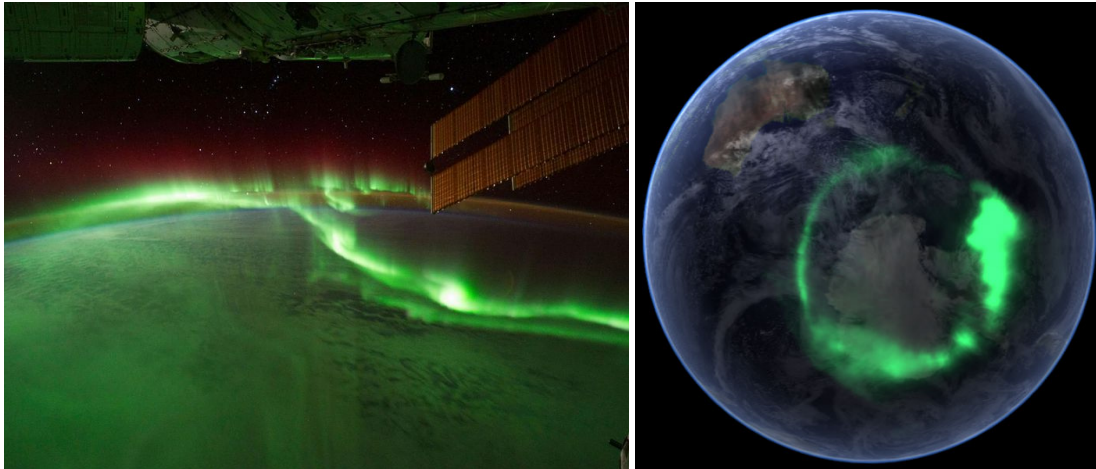


Figure 21: The aurora, as viewed (left) from low-Earth-orbit (the space station) and (right) from far below the south pole, showing the auroral oval.

4.3 Navigation

The Earth's magnetic field has been used for navigation for thousands of years by human and millions of years by animals; the first compass was invented in China between 300 and 200 BC. In the 21st century, one might ask if we still need to use compasses for navigation, if we have GPS?

The answer is yes. GPS (and its predecessor, radio navigation) is very good at providing *position*, but does not provide any information on *bearing*, i.e., what direction you are pointing or moving. Here on the Earth's surface, or in space, a compass or *magnetometer* is still the best and easiest way to determine the bearing.

But, magnetic navigation is only useful if you know where the compass is pointing, and that requires knowledge of the *magnetic declination* anywhere on the planet⁴. The WMM provides this information in its model updated every five years; the declinations don't change appreciably on shorter time scales.

During geomagnetic storms, the magnetic fields at Earth's surface can change, but usually less than 1%. These changes are also temporary, lasting just a few days (see Figure 20 above), so the field will soon return to its ambient configuration. For these reasons, geomagnetic storms are not a major concern for magnetic navigation on the surface of the Earth. However, there are observations that pigeons' homing success is reduced on days with high geomagnetic disturbance.

⁴A common misconception is that the compass points towards magnetic north: this is incorrect! A compass points along the horizontal component of the local geomagnetic field, which generally differs from true north by an angle called magnetic declination. It does not necessarily point directly toward the magnetic North Pole.

Spacecraft also use magnetic navigation; it is common for the spacecraft Attitude Determination and Control System (ADCS) to incorporate a magnetometer, particularly in LEO. In this case, the magnetometer can determine the magnetic field magnitude and direction, but the direction in particular is useful to provide a relative orientation (attitude) of the spacecraft. However, to be useful, the system must know the direction of the magnetic field locally. The magnetometer is usually paired with a GPS receiver for position, and then it uses the position and the magnetic field model (WMM or IGRF) to determine the local magnetic field direction. The spacecraft orientation can then be determined in Earth-centered coordinates.

The attitude control system of Colorado Student Space Weather Experiment (CSSWE) is entirely passive, using bar magnet and hysteresis rods to roughly align the long axis of the spacecraft with the background magnetic field. The attitude with respect to Earth's magnetic field is known after analyzing the housekeeping data. REPTile's look direction is, for the vast majority of the time, close to 90 deg with respect to the background magnetic field and thus measuring locally mirroring particles. However, as the field of view of the instrument is large (plus/minus 26 deg), the measured particles are a combination of trapped and precipitating populations, depending the location of the measurements, e.g., dominated by trapped population in SAA.

4.4 Geomagnetically Induced Currents

We saw above that geomagnetic storms drive enhancements in the ring current, as well as in other particle populations (i.e., the plasmasheet). They also drive enhancements in the other currents we described earlier: field-aligned currents, the neutral current sheet, and so forth.

In the ionosphere module we introduced the **Equatorial Electrojet**; another current system in the ionosphere is the *Auroral Electrojet*. Currents also flow in other latitudes of the ionosphere, driven by electric fields and winds. During geomagnetic storms, all of these ionospheric currents can be perturbed and/or enhanced dramatically. These enhanced currents drive magnetic fields due to the Biot-Savart law; such a magnetic field is measured at the Earth's surface as a perturbation to the background magnetic field, similar to the Ring Current Dst effect. Importantly, the ionosphere currents are much closer to the Earth, and can thus drive strong magnetic fields. These fields in turn lead to currents in the conducting Earth surface – these are called **Geomagnetically Induced Currents** or GICs.

Now, the *current density* created in the Earth's surface is not huge. It also depends on the surface conductivity, which varies by orders of magnitude across the planet⁵. Because of the finite conductivity in the Earth, those currents would much rather find a low-resistance path to flow through. Humanity now provides those paths: through long-distance power lines, pipelines, or railroads on the surface.

During large storms, GICs of tens of Amperes have been measured through power lines. This is problematic for power lines: they are designed to operate at high voltage and low current, so these large pulses of high current can be quite disruptive.

The effects of GICs on the three structures mentioned above are:

- **Power lines:** The large pulse of current causes a reduction in the power line voltage. The operation of high-voltage power lines is such that if the voltage drops below some minimum, it can collapse to zero voltage, breakers will trip, and the system will shut down. This leads to blackouts. This is the same way blackouts occur due to high loads on power systems: high loads pull large currents, which lead to voltage drop (brownouts) or shutdown (blackouts).
- **Pipelines:** Underground pipelines are typically held to a negative voltage to prevent corrosion (or prevent the loss of electrons from the pipes). During GICs, the pipe-to-soil voltage can change dra-

⁵The conductivity is highest in the oceans, orders of magnitude higher than the ground, but we'll ignore the oceans for now.

matically, increasing corrosion during major storms. The concern is not of sudden pipeline failure, but of increased wear over time.

- **Railroads:** This one is interesting: railroad switches are designed to operate automatically. The railroad itself is part of a circuit; when a train is present, the conducting wheels close the circuit and allow current to flow. This current is sensed by the switches and allows them to open or close. During GICs, the switches can be confused into switching their state in the absence of a train. The ground-induced currents (GICs) from magnetic storms have caused railway signal malfunctions.

GIC effects are observed during every major geomagnetic storm, but a couple from the past stand out. In March 1989 a large storm with $Dst < -600$ nT caused a massive blackout in Quebec: over 6 million people lost power for over 9 hours. Brownouts were observed in the Northeastern US as well. Quebec was hit hard because the [Canadian Shield](#) (a big slab of rock covering half of the country) is a very poor conductor, so the currents in the power lines were extremely strong.

That magnetic storm pales in comparison to the massive **Carrington Event** in September 1859. The Dst index reached -1600 nT. Long distance power lines were not so common at this time, but telegraph lines were. Systems failed all over the US and Europe, in some cases with shocks or even fires. Auroras were observed as far south as the Caribbean and Mexico.

A similar event occurred in 2012 that missed the Earth by just 9 days (in orbit phase; considering the Earth's orbit is 365 days, that's about 9 degrees of phase). Had it hit the Earth, it has been estimated that such a storm would cause damages of \$1–3 trillion (not a typo!). Events like this are expected to occur at Earth about once every 500 years on average, but these massive storms are part of the reason we study space weather, in order to better predict, forecast, and mitigate the effects of such storms.

4.5 Spacecraft Charging (Again)

Recall in the Ionosphere module we talked about spacecraft charging, both in LEO and GEO. We found that in LEO, the spacecraft floating potential, V_{fl} , is generally small, on the order of Volts. At GEO, however, the spacecraft potential can be many thousands of volts. The equation for spacecraft floating potential was given by

$$V_{fl} = \frac{k_B T_e}{q_e} \ln \left[\sqrt{\frac{m_e}{m_i}} \left(1 - \frac{q_e V_{fl}}{k_B T_i} \right) \right] \quad (35)$$

which unfortunately has no closed form solution. The approximate solution that is often used is given by

$$V_{fl} \simeq -2.5 T_e \quad (36)$$

with T_e given in eV. We found that for an electron temperature of 10 million Kelvins (~ 1 keV), the floating potential is about -2500 V.

Where did this 10 million Kelvin temperature come from? The plasmasphere is not nearly that hot (~ 1 eV is equivalent to about 11,000 K), but the plasma sheet and ring current ions and electrons can reach these very high temperatures. The electron temperature T_e of the plasma sheet⁶ varies from 2–20 million Kelvins, and so the spacecraft potential can similarly vary by an order of magnitude.

⁶The ring current has an average energy of 85 keV, which using our normal conversion, would correspond to a temperature of 10^9 K; however the ring current is not in a Maxwell-Boltzmann distribution, so this conversion is not really valid.

5 Missions

5.1 Magnetic Field Missions

The Ørsted, CHAMP, and Swarm missions have a number of science objectives, but common to all three is the precise measurement of Earth's magnetic field from space. These missions use precise magnetometers to measure both the **scalar** field (that is, the magnitude of the magnetic field) and the **vector** field (the magnitude of each of the three components).

You might ask: why do you need a separate scalar measurement if you have all three vector components? The type of magnetometer used to measure the vector field, known as a vector fluxgate magnetometer (VFM), can measure the three components with high resolution, but they are typically susceptible to drift and offset; as such, the calculated field magnitude from a VFM may have an error of 100 nT or more. So, the *scalar* magnetometer is used to measure the precise magnitude, which can then be used to correct the VFM measurements.

On all of these spacecraft, the magnetometers are installed on the end of a long boom – 8 meters long for Ørsted and 3–5 m for the others (Figure 22). The boom is required to isolate the magnetometer from any magnetic noise or bias produced by the spacecraft and its electronic systems. Recall that currents produce magnetic fields; electronics tend to have currents running through them. In addition, any magnetic materials (iron, etc) in the spacecraft may have residual magnetic fields. Placing the magnetometer far from the spacecraft structure and magnetic materials ensures that these fields do not corrupt the desired measurements.

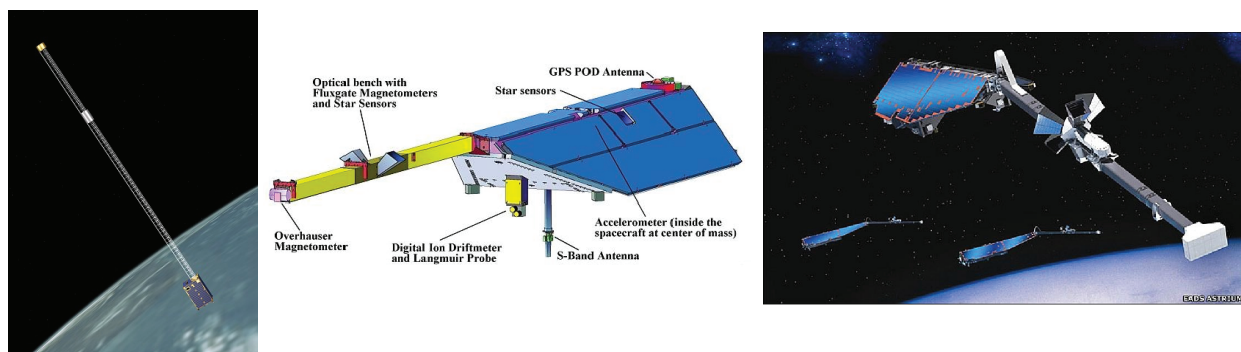


Figure 22: Left to right: Ørsted, CHAMP, and SWARM missions.

The magnetic field measurements from these spacecraft are critical to the global magnetic field models, WMM and IGRF. While most of the model data comes from ground-based magnetometers, the space-based measurements fill in the holes where there are no magnetometers (i.e. over the oceans, antarctica and the arctic, etc.), and help to map fine-scale features of the magnetic field created by deposits of iron in the Earth's crust: so-called “crustal fields”.

Read more about these missions here:

Ørsted: <https://earth.esa.int/web/eoportal/satellite-missions/o/oersted>

CHAMP: <https://directory.eoportal.org/web/eoportal/satellite-missions/c-missions/champ>

Swarm: <https://directory.eoportal.org/web/eoportal/satellite-missions/s/swarm>

three ESA satellites (A, B, C), launched in 2013; all three remain in orbit and the mission is still operational (2026). There is currently no plan to replace SWARM for the necessary magnetic field measurements for the WMM model. Professor Marshall is currently leading a CubeSat mission (COSMOS: Compact Spaceborne Magnetic Observatory) to make these measurements, COSMOS was launched in early 2026 and is currently in commissioning phase.

5.2 Magnetospheric Multiscale (MMS)

The Magnetospheric Multiscale (MMS) Mission was designed for one main purpose: to understand the physics of “reconnection”, the process when two oppositely-aligned magnetic fields come together and “break” into a new configuration (Figure 23). The two opposite-polarity magnetic fields are being dragged together by flowing plasma (i.e., from the solar wind); at the point of reconnection, the physics describing what happens to this plasma is very complicated. It’s a difficult phenomenon to observe in a laboratory environment due to the very small scale sizes present; as such, taking the measurements out into space, where scale sizes are considerably larger, makes the problem somewhat easier. In this way MMS isn’t just a space physics mission, it is a mission to investigate an important aspect of fundamental plasma physics.

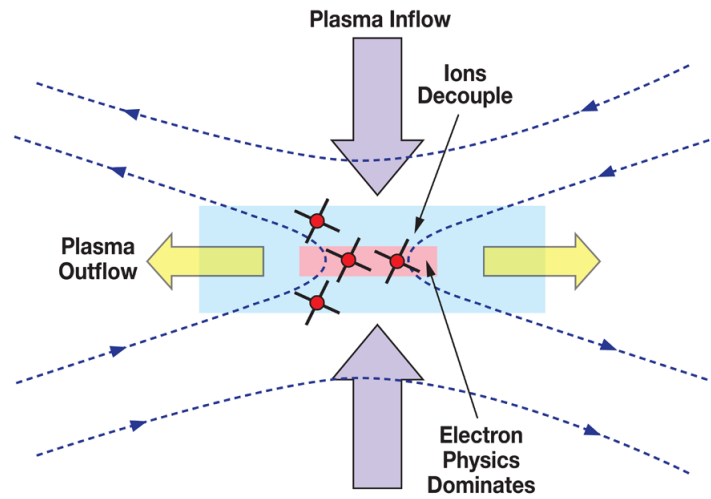


Figure 23: Magnetic Reconnection and MMS.

Read more about MMS here:

EO Portal: <https://directory.eoportal.org/web/eoportal/satellite-missions/m/mms-observatory>

NASA: <https://mms.gsfc.nasa.gov>

Published Literature: <https://link.springer.com/article/10.1007/s11214-015-0164-9>

6 Movie: The Core

Synopsis from Fandango Media:

Geophysicist Dr. Josh Keyes (Aaron Eckhart) discovers that an unknown force has caused the earth’s inner core to stop rotating. With the planet’s magnetic field rapidly deteriorating, our atmosphere literally⁷ starts to come apart at the seams with catastrophic consequences. To resolve the crisis, Keyes, along with a team of the world’s most gifted scientists, travel into the earth’s core. Their mission: detonate a device that will reactivate the core.

The Core is arguably⁸ the worst mainstream space movie ever made, and a favorite among space scientists – mention it at any conference and get a knowing, mocking smile (we can be a smarmy bunch). It has a strong cast and a big budget, and it’s well made, up until the terrible special effects inside the Earth’s core.

⁷ And by “literally” the writer obviously means “figuratively”.

⁸ *Armageddon* and *Deep Impact*, of similar vintage, are also up there on the list.

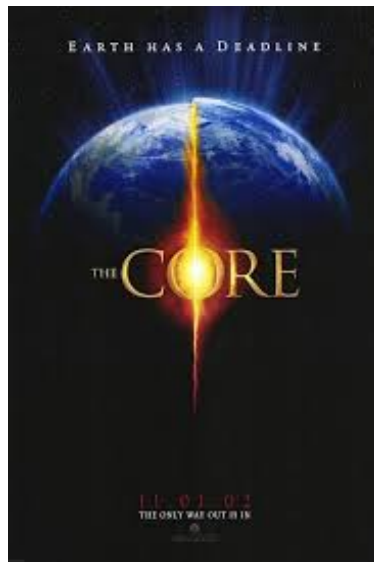


Figure 24: The Core (2003).

It seems that any movie requires a plausible premise if it's going to succeed. The basic idea, that the Earth's magnetic field protects us from the sun, is a terrible extrapolation of a moderately accurate idea. Based on our discussions of the atmosphere earlier and the magnetosphere in this module, you should be able to figure out what's wrong with this premise.

Problems

1. **(10) Field Magnitudes:** Using this [online calculator](#), determine the magnetic field in Boulder, CO, and at McMurdo Station, Antarctica. (You can use either the IGRF or WMM model, but let us know which one you use!) Report the total field magnitude, the declination, and the inclination. What does the inclination at McMurdo tell you about the magnetic field lines above this region?
2. **(10) Dipole Field:** Assuming a dipole field model, what is the magnetic field magnitude at $L = 4$, i) at the equator, and ii) at $\lambda = 30^\circ$?
3. **(10) Magnetopause:** The magnetopause is the boundary where the solar wind pressure equals the magnetic pressure of the magnetosphere.
 - i) Assuming a dipole field with $B_0 = 31,000$ nT, and a solar wind moving at 400 km/s with a density of 5 protons/cm³, estimate the standoff distance, i.e., the distance from the center of the Earth to the magnetopause.
 - ii) During a particular CME, the solar wind increases to $v = 1000$ km/s and the proton number density doubles (these changes are typical, but it can vary!). What is the standoff distance under these conditions?
4. **(10) The Ring Current:** Ring current particles have an average energy of 85 keV. Consider the case where they have only $v_\perp$, i.e. $v_\parallel = 0$, and their motion is strictly in the equatorial plane. For this problem, consider the effects of the magnetic field gradient only.
 - i) Explain why there is a current in the first place. (There are two key factors.)
 - ii) How long does it take an 85 keV electron to drift all the way around the Earth at $L = 4$?
5. **(10) Drift due to Gravity:** As above, consider charged particles near the equator at $L = 4$.
 - i) Under the influence of gravity, what direction will protons and electrons drift?
 - ii) What is the magnitude of the drift velocity due to gravity? (Don't forget that gravity is weaker as we move further from the Earth.)
 - iii) How long will it take particles to drift around the Earth under gravity drift? Compare your answer to Problem 4 above. Should we include gravity drift in our ring current calculations?
6. **(10) Geomagnetic indices:** In this problem, we'll investigate the distribution of the Kp index over a 50-year period.
 - go to <https://omniweb.gsfc.nasa.gov/ow.html>
 - Click on "Distributions functions, averages and std deviations" in the top left yellow box.
 - Click the radio button for "Get distribution function etc"
 - Enter the start date as 19640101 and the end date as 20161231
 - Click the radio button for Kp*10 Index (this will be the numerical version of Kp multiplied by 10)
 - Further down on the page specify the bin size as 1
 - Click Submit
 - Include the figure in your assignment submission.

Now answer these questions:

- i) What is the most likely value of K_p ?
- ii) What fraction of the K_p values is observed to be at level 9? (hint: click on “Listing of Distribution Function table” above the image and use the data to calculate the necessary statistic.)
- iii) Repeat the procedure above to get a histogram of the K_p data for the month of July 2004. Compare the relative frequency of the $K_p > 5$ for this interval to the multi-year interval above. Would you say that July 2004 was more or less disturbed than typical?

Note that you can use this online tool to determine the typical values and distribution functions of many solar wind parameters, energetic particles fluxes, geomagnetic indices, and more.

7. **(10) Geomagnetic Storm:** In addition to OmniWeb above, historical Dst index data is available from this website: <http://wdc.kugi.kyoto-u.ac.jp/dst/dir/index.html>.

- i) Use this site to create a figure showing the Dst index for the month of October 2003. (The event shown is known as the “Halloween Storm.”)
- ii) Calculate the ring current magnitude (in Amperes) necessary to create the observed Dst of -400 nT. Assume the ring current can be modeled as a circular loop of current at $5R_E$.

8. **(10) Key Parameters in the Solar Wind Driving a Magnetic Storm**

- i) Among the parameters describing solar wind: velocity, plasma density, and interplanetary magnetic field, which parameter is the most critical one, in what way? to determine if a magnetic storm will occur? Why?
- ii) Look at the Dst index profile for a magnetic storm (you should look through a few of them to convince yourself), you will see a certain surge (a positive change) in Dst index before the Dst index decreases (becomes more negative) as the main phase of the storm starts. What kind of current likely causes the initial surge of Dst index before the main phase of the magnetic storm?

9. **(10) The Core:** Just based on the 3-minute clip posted on Canvas, list i) three statements about the space environment that are correct, and ii) three statements about the space environment that are complete garbage.

10. **(10) Magnetospheric Multiscale:**

- i) What is the primary objective of the MSS mission?
- ii) Why does the mission require four spacecraft? What arrangement of the four spacecraft is used?
- iii) What instruments are on each of the four spacecraft? How many of each? Why are there multiples of some instruments? (hint: the spacecraft are *spin-stabilized*; what does that mean?)
- iv) What are the two main orbit phases of the MMS mission, and why? What are the apogee and perigee of each phase?